

# DATA CO GLOBAL SUPPLY CHAIN ANALYSIS

Focus on Sales and Delivery Quality



# Presentation Orders

1. Analysis goal and plan
2. About the datasets
3. Data Cleaning
4. EDA process - Sales examination
  - a. Refine sales data
  - b. Main KPI for sales
  - c. Sales trend and related factors
5. Late Delivery examination
6. Insight - Recommendations



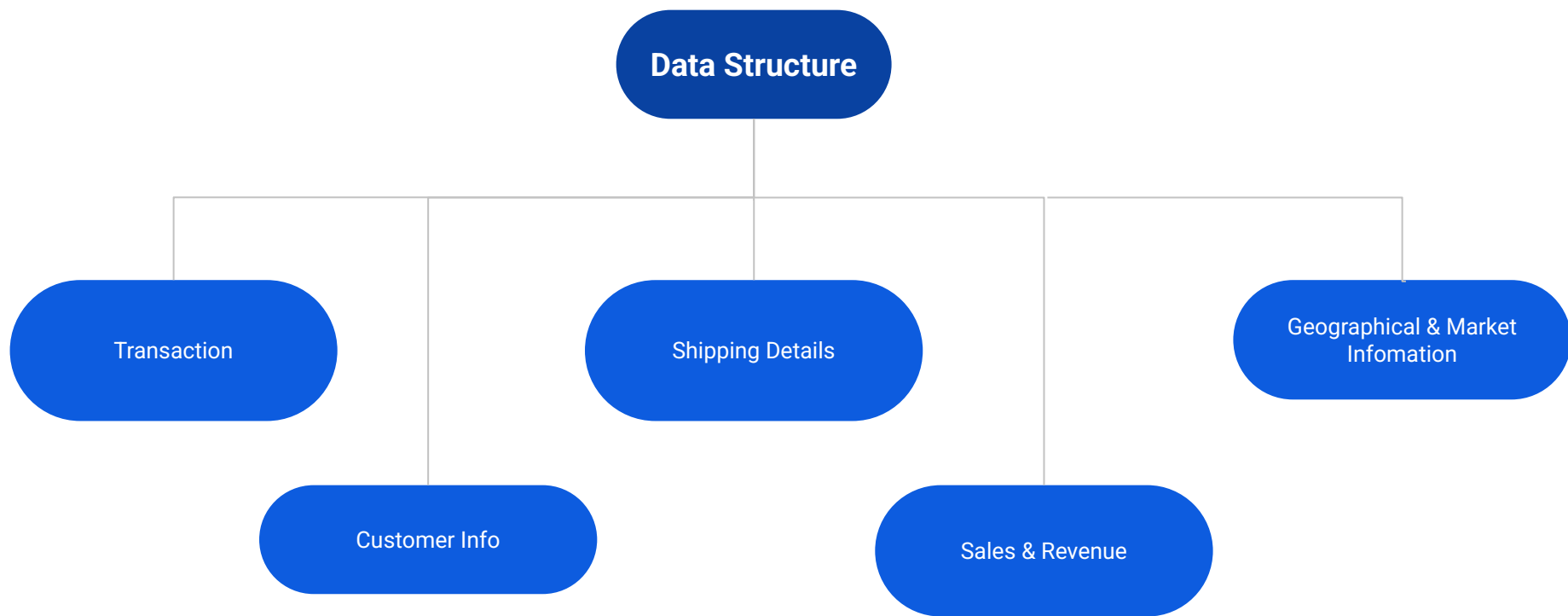
# Analysis Goal & Plan

- Goal : making a **descriptive analyst** on DataCo SMART supply chain datasets to find insights and places to improve (main focus will be **Sales and Delivery Quality**)
- Plan



# About the dataset

Dataset is a stimulated Global Retailer by DataCo SMART to analysis their supply chain



# Data Cleaning

# Data Cleaning

## Overview

```
[3] #Start with checking for data info
df.info()
```

<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 180519 entries, 0 to 180518  
Data columns (total 53 columns):

| #  | Column                        | Non-Null Count  | Dtype   |
|----|-------------------------------|-----------------|---------|
| 0  | Type                          | 180519 non-null | object  |
| 1  | Days for shipping (real)      | 180519 non-null | int64   |
| 2  | Days for shipment (scheduled) | 180519 non-null | int64   |
| 3  | Benefit per order             | 180519 non-null | float64 |
| 4  | Sales per customer            | 180519 non-null | float64 |
| 5  | Delivery Status               | 180519 non-null | object  |
| 6  | Late_delivery_risk            | 180519 non-null | int64   |
| 7  | Category Id                   | 180519 non-null | int64   |
| 8  | Category Name                 | 180519 non-null | object  |
| 9  | Customer City                 | 180519 non-null | object  |
| 10 | Customer Country              | 180519 non-null | object  |
| 11 | Customer Email                | 180519 non-null | object  |
| 12 | Customer Fname                | 180519 non-null | object  |
| 13 | Customer Id                   | 180519 non-null | int64   |
| 14 | Customer Lname                | 180519 non-null | object  |
| 15 | Customer Phone                | 180519 non-null | object  |
| 16 | Customer Zip                  | 180519 non-null | object  |
| 17 | Product Id                    | 180519 non-null | int64   |
| 18 | Product Name                  | 180519 non-null | object  |
| 19 | Product Category Id           | 180519 non-null | int64   |
| 20 | Product Description           | 0 non-null      | float64 |
| 21 | Product Image                 | 180519 non-null | object  |

- Some columns are either blank or missing value
- dtype should be converted from object to category/string
- Remove duplicates and privacy info columns

# Data Cleaning

## Process Brief

### Remove columns

#### Unnecessary columns to be removed

Customer private info  
(name, address, email  
etc.)

Link, product description  
which unanalyzable

Columns contain same  
info

### Convert dtypes

#### Object to string/category/datetime

City name,

Customer segment

Product categories

Date of order/shipping

etc.

### Fill missing value & Remove duplicates

#### Fill na & check for duplicates

Some missing values has  
been removed

No duplicates records  
was found

# Data Cleaning

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180519 entries, 0 to 180518
Data columns (total 34 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Type                                  180519 non-null category
1   Days for shipping (real)              180519 non-null int64
2   Days for shipment (scheduled)         180519 non-null int64
3   Sales per customer                    180519 non-null float64
4   Delivery Status                       180519 non-null category
5   Late_delivery_risk                    180519 non-null int64
6   Category Name                         180519 non-null category
7   Customer City                         180519 non-null string
8   Customer Country                     180519 non-null string
9   Customer Id                           180519 non-null int64
10  Customer Segment                      180519 non-null category
11  Customer State                        180519 non-null string
12  Department Name                       180519 non-null string
13  Latitude                              180519 non-null float64
14  Longitude                             180519 non-null float64
15  Market                               180519 non-null string
16  Order City                            180519 non-null string
17  Order Country                         180519 non-null string
18  Order Id                              180519 non-null int64
19  Order Item Discount                   180519 non-null float64
20  Order Item Discount Rate              180519 non-null float64
21  Order Item Profit Ratio               180519 non-null float64
22  Order Item Quantity                   180519 non-null int64
23  Sales                                 180519 non-null float64
24  Order Item Total                      180519 non-null float64
25  Order Profit Per Order                180519 non-null float64
```

```
26  Order Region                          180519 non-null string
27  Order State                           180519 non-null string
28  Order Status                          180519 non-null category
29  Product Name                          180519 non-null string
30  Product Price                         180519 non-null float64
31  Shipping Mode                         180519 non-null category
32  DateOrders                            180519 non-null datetime64[ns]
33  DateShipping                          180519 non-null datetime64[ns]
dtypes: category(6), datetime64[ns](2), float64(10), int64(6), string(10)
memory usage: 39.6 MB
```

## (Results)

- Removed unnecessary columns (52 -> 33 columns)
- All categorized, string or datetime converted
- None missing or duplicate values (180519 records)



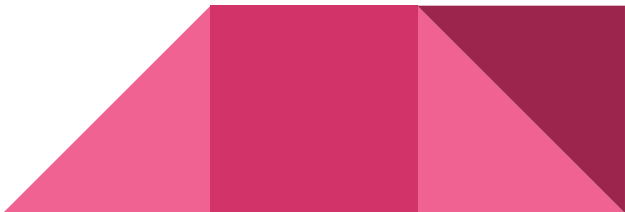
# Exploratory Data Analysis

# Refine sales data

Sales contain records with order status 'CANCELED', 'ON\_HOLD' and 'SUSPECTED\_FRAUD' will be replaced by 0

```
#adding a columns name refined_sales. When order status are" canceled, on_hold or suspected_fraud will be 0, the rest remain original value.
df['refined_sales'] = df['Sales']

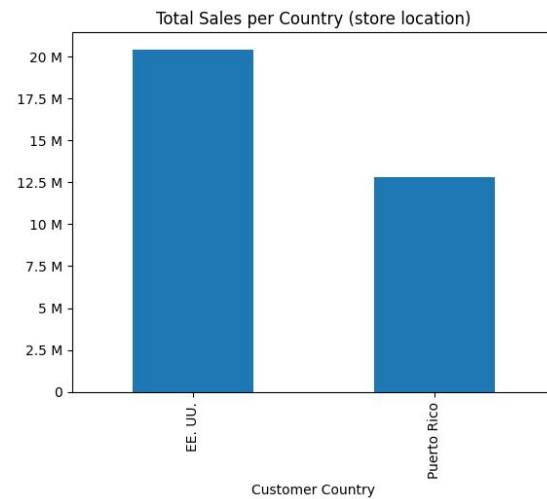
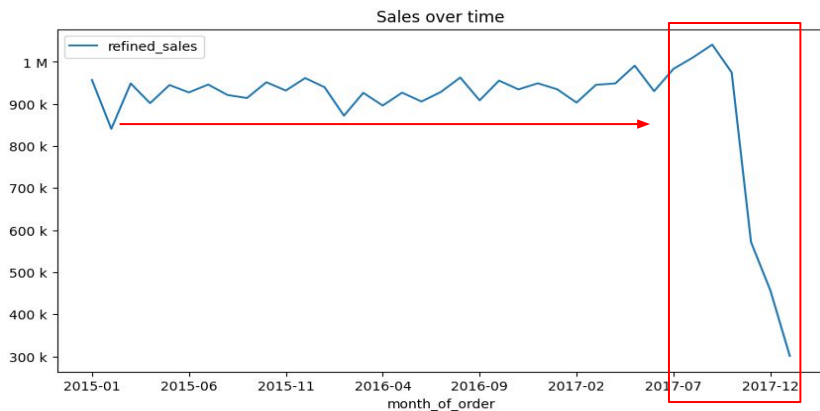
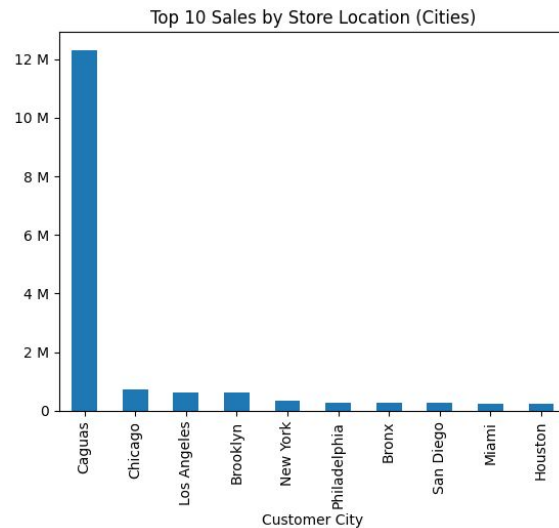
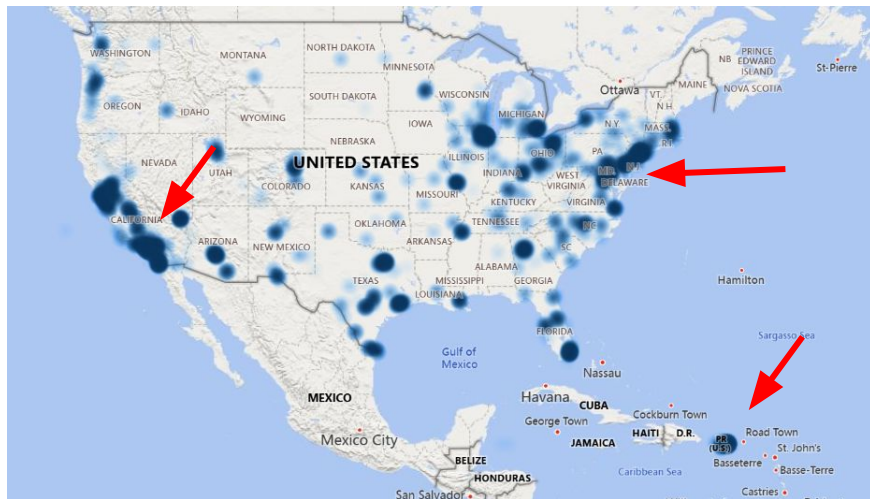
# Replace values in 'refined_sales' with '0' where 'Order Status' is in canceled, on_hold or suspected_fraud
df.loc[df['Order Status'].isin(['CANCELED', 'ON_HOLD', 'SUSPECTED_FRAUD']), 'refined_sales'] = 0
```



# MAIN KPI

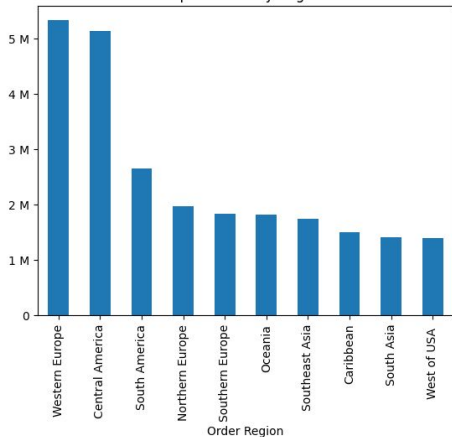
- Total Order: 65752
  - Total Sales 33.23M
  - Average Sale per Order: 505.43
  - Average Sale per Month: 898186.13
  - **Late Delivery Rate: 54.83%**
  - Average Discount Rate: 10.17%
  - Total Quantity Sold: 384k
  - Total Customer: 20,652
  - Repeat Customer Rate: 55.38%
  - The most profitable product: Field & Stream Sportsman 16 Gun Fire Safe
  - The Top Region by Sale: Western Europe
- 

# Sales trend and related factors

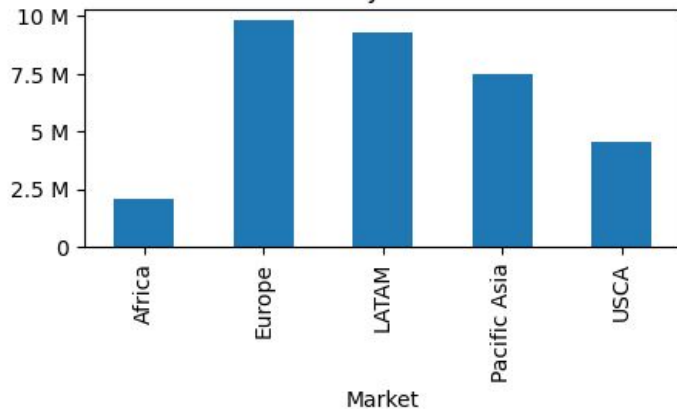


# Sales trend and related factors

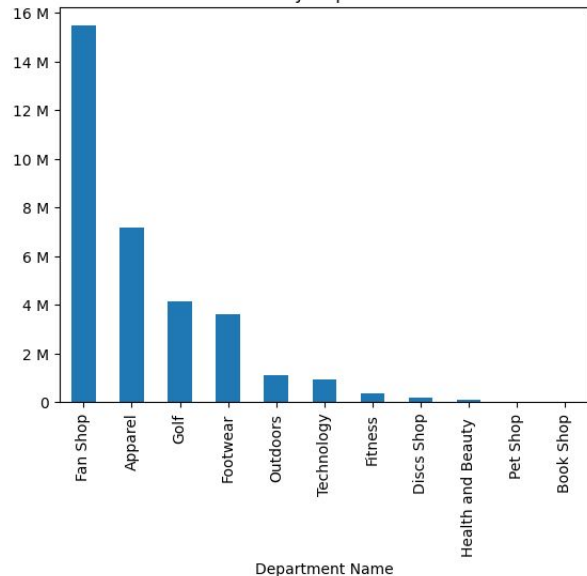
Top 10 Sales by Region



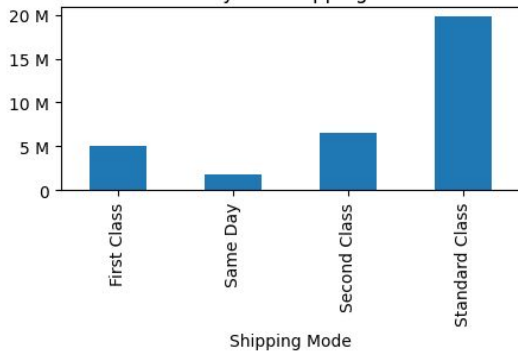
Sale by Market



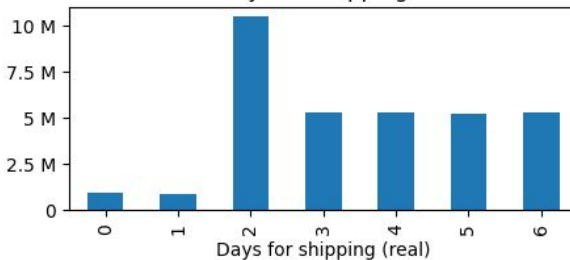
Sales by Department



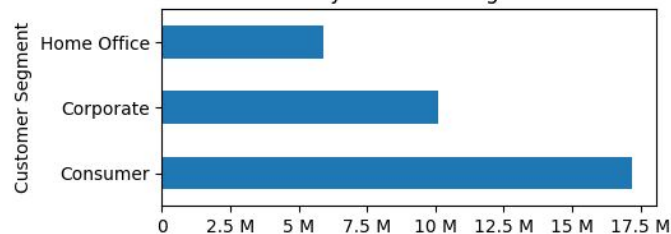
Sale by real shipping time



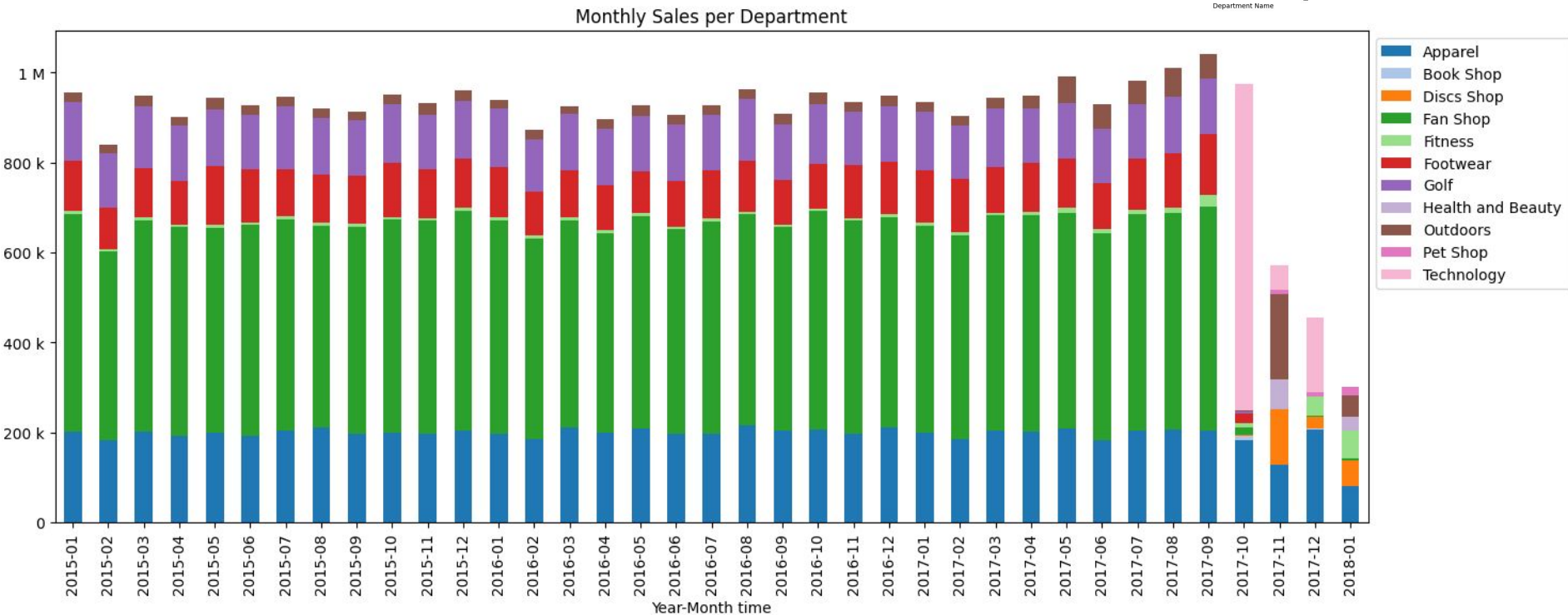
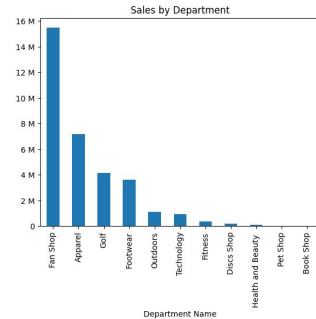
Sale by real shipping time



Sale by Customer Segment

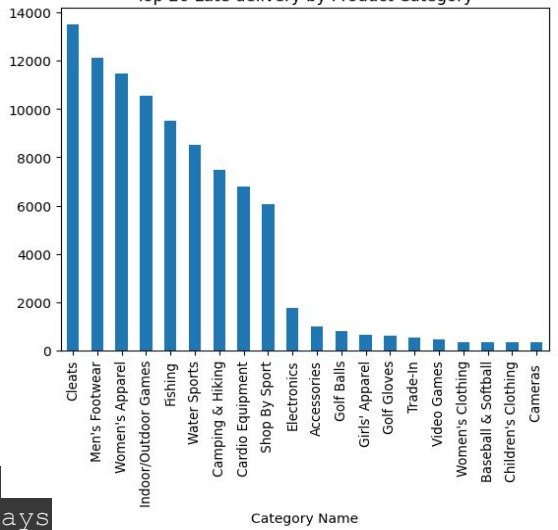
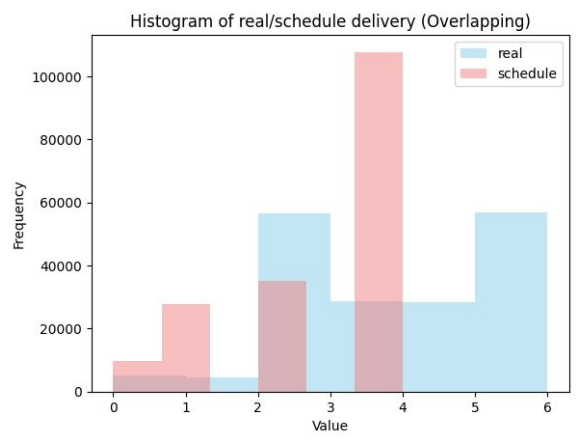
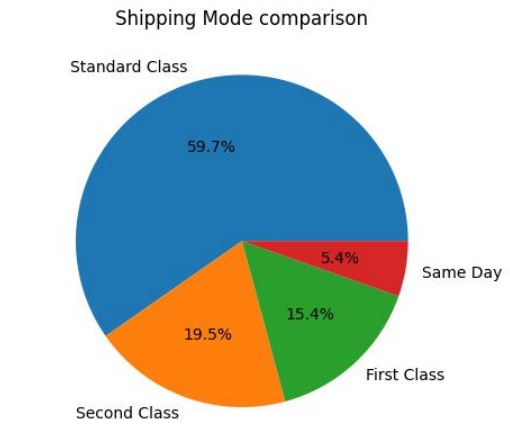


# Sales trend and related factors

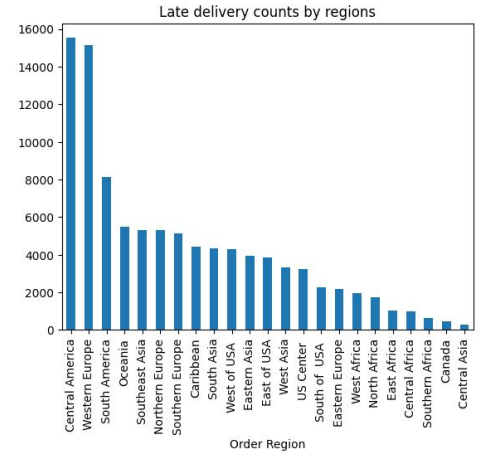
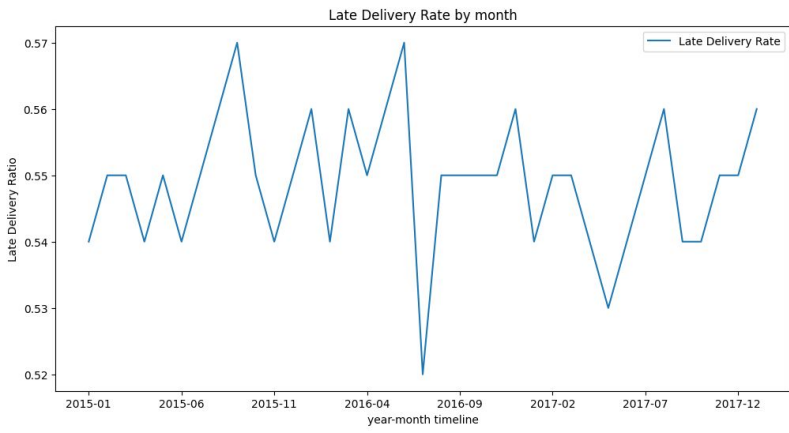
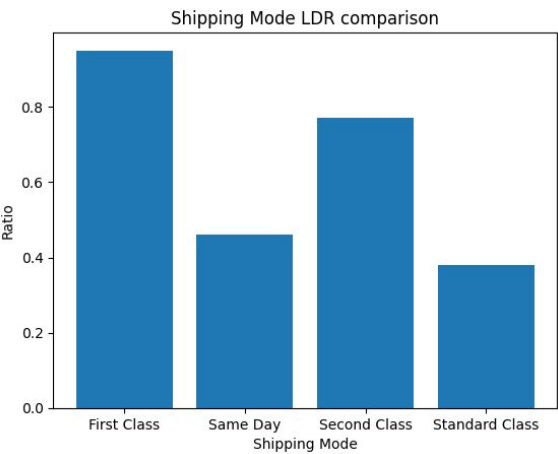


# Late Delivery examination

# Late delivery inspect



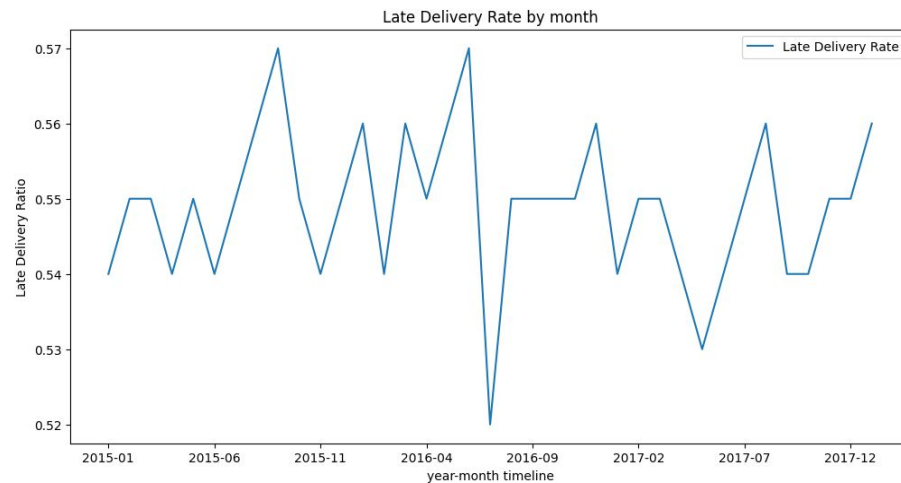
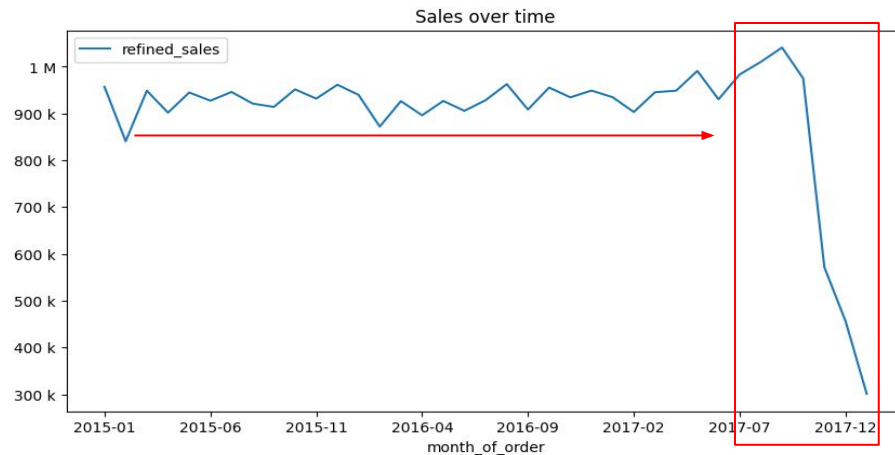
Average process time when late delivery: 4 days  
Average process time when on-time delivery: 2 days





# Noticeable Problems

- Dropping Sales since 2017 Oct
- Extended late delivery



# The most possible causes for each issue:

- For dropping sales:

The sales are drop since 2017-Oct due to the **disappear of many sale-leading department** such as fan shop, apparel, etc. Instead, they getting sales from new department (since 2017-Oct) such as Technology, Health and Beauty, Fitness only.

- For high Late Delivery Rate:

The rate remain fluctuating between 52% and 56% during 2015-2017 and seems to be increasing in late 2017.

Late Delivery Rate is affected by many factors. In the scope of our available data, we may point out these causes:

- **Long processing time** from order to actual shipping (4 days instead of the required 2 days)
- Some **special products** require more shipping time than others (such as Cleats, Men's footwear, Women's apparel etc.)
- Shipping Classes have been **under-performing**. Especially First and Second Classed are often late while they are supposed to be on-time or advanced.
- The schedule **shipping time is not precise**. Especially for 1-2 days delivery orders. We need to improve the precision and should consider more about actual average shipping time 3.5 days.



# Action plan

- For sales:

Because the limited of available data, we can't make conclusion on the disappear of leading sales department like fan shop and apparel. If we can **bring back the sale-leading departments**, the total number will recover.

- For delivery:

- We should focus more on **improving Average Order Process Time** from 4 days down to **2 days**.
- We should re-qualifying and **improve the quality of premium shipping modes** (First Class and Second Class).
- We should start **optimizing delivery routes** to Central America, South America and Western Europe.
- In addition, schedule shipping time also needs to be **well considered before inputting**.
- Finally, be **well prepared around the end of year** period since it often has more late delivery count than others.

